

EKTA PATEL

University of Arizona
Steward Observatory
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website: [ekta1224.github.io](https://github.com/ekta1224)
(US Citizen)

Research interests: Near-field cosmology, satellite galaxies, dwarf galaxies, computational astrophysics

EDUCATION

University of Arizona, Tucson, AZ

Ph.D. in Astronomy & Astrophysics

expected May 2019

M.S. in Astronomy

May 2017

Advisor: Dr. Gurtina Besla

New York University, New York, NY

B.A. in Physics (with Honors) and minors in Astronomy and Mathematics

May 2014

Senior Honors Thesis: *The Sloan Digital Sky Survey Large Galaxy Atlas*

Advisor: Dr. David Hogg

FELLOWSHIPS, HONORS, AND AWARDS

- 2018: Academy of Tucson High School Keynote Speaker
- 2016 - 2019: National Science Foundation Graduate Research Fellowship Recipient
- 2016: University of Arizona College of Science Service Award Recipient
- 2015: Ford Foundation Pre-Doctoral Fellowship Honorable Mention and Alternate Candidate
- 2012: NYU Dean's Undergraduate Research Fund Recipient
- 2011 - 2014: NYU Presidential Honors Scholar
- 2011 - 2014: NYU Women in Science Scholar and Marilyn Miles Scholarship Recipient

RESEARCH EXPERIENCE

University of Arizona, Steward Observatory

Tucson, AZ

Graduate Research Assistant, Advisor: Dr. Gurtina Besla

September 2014 - Present

- My doctoral thesis focuses on the evolution and dynamics of satellite galaxies in the Local Group within a theoretical framework.

New York University, Center for Cosmology and Particle Physics

New York, NY

Undergraduate Research Assistant, Advisor: Dr. David Hogg

March 2012 - July 2014

- I worked with large data sets including *SDSS* and the *GALEX* photon catalog to improve photometry for galaxies with large angular sizes and to successfully find the source of systematically noisy data.

Max-Planck-Institut für Astronomie

Heidelberg, Germany

Visiting Research Assistant

July - August 2013, 2014

Advisors: Dr. David Hogg, Dr. Hans-Walter Rix

- I helped develop probabilistic models for inferring the photometry of time-variable quasars in *SDSS Stripe 82* and created a model to search for potential photon flares in the *GALEX* photon catalog.

CONFERENCES AND TALKS

Invited: 6

- Astro Lunch Seminar, University of Pittsburgh/Carnegie Mellon University, Spring 2019
- CCAPP Seminar, Ohio State University, November 2018
- Department Lunch Talk, UC Berkeley Astronomy Department, October 2018
- Astrophysics Seminar, American Museum of Natural History Astronomy Department, October 2018
- PHAT Collaboration Team Meeting, Ringberg Castle, July 2018
- 232nd AAS Meeting Press Conference, Denver, CO, June 2018

Contributed: 6

- 232nd AAS Meeting, Milky Way Session 402, Denver, CO, June 2018
- Large Surveys of the Great Andromeda Galaxy, Lorentz Center, July 2017
- Marc Aaronson Symposium, University of Arizona, April 2017
- Local Group Astrostatistics Conference, University of Michigan, poster presentation, June 2015
- Midwest Conference for Undergraduate Women in Physics, University of Chicago, poster presentation, January 2014
- Dean's Undergraduate Conference, New York University, poster presentation, April 2013, *Best in team poster presentation*

Other: 8

- ExGal Seminar, University of Texas at Austin, April 2018
- JILA Seminar, JILA Institute/University of Colorado, December 2017
- Center for Computational Astrophysics Stars Group Meeting, Flatiron Institute, October 2017
- Galaxies Lunch Seminar, Columbia University, October 2017
- Galaxy Club, Space Telescope Science Institute, December 2016
- FLASH Talk, National Optical Astronomy Observatory, November 2016
- Hernquist Group Meeting, Harvard-Smithsonian Center for Astrophysics, September 2016
- Journal Club, University of Arizona, April 2016

PROFESSIONAL EXPERIENCE

Computational & Software Development

Experience with Python, C, Emacs, and Git (<https://github.com/ekta1224>)

- 2016: Python in Astronomy Workshop, University of Washington
- 2013: Scicoder Workshop, American Museum of Natural History

Grant Writing

- 2017: *Editor* for the University of Arizona Graduate College Application Support Program, a program that aids graduate students with their pre-doctoral, graduate, and dissertation fellowship applications

Academic Service

- Referee for *The Astrophysical Journal*
- 2017: Steward Observatory Prospective Graduate Student Visit Co-Organizer (4 days, 17 students)
- 2016: NASA Astrophysics Theory Program Executive Secretary
- 2015 - 2017: Steward Observatory Graduate Council Member
- 2013 - 2014: NYU Society of Physics Students President & Outreach Coordinator

COMMUNITY ACTIVITIES

Public Presentations

- 2018: *Galactic Archaeology: From Little to Big*, NOAO Teen Astronomy Cafe
- 2018: *Satellite Galaxies and Dwarfs in the Local Group*, Sonora Astronomical Society
- 2017: *Satellite Galaxies and Dwarfs in the Local Group*, Tucson Amateur Astronomy Association

Engagement

- 2018: *Colors of Nature Summer Program Instructor*, Kitt Peak, AZ
- 2018: *Colors of Nature Summer Program Instructor*, Arizona-Sonora Desert Museum, Tucson, AZ
- 2017: *St. Michael's School 8th Grade Science Fair Judge*, Tucson, AZ
- 2017: *Colors of Nature Summer Program Instructor*, Arizona-Sonora Desert Museum, Tucson, AZ
- 2016 - 2017: *Classroom Astronomer*, 9th Grade City High School, Tucson, AZ
- 2015: *Fruchthendler Elementary School Science Fair Judge*, Tucson, AZ
- 2015: *Senita Valley Elementary School Family Science Night Volunteer*, Tucson AZ
- 2015 - 2016: *Project ASTRO Classroom Astronomer*, 5th Grade Cragin Elementary School, Tucson, AZ

Mentoring

- 2015 - Present: *Tucson Initiative for Minority Engagement in Science and TEchnology Program (TIMESTEP) Mentor*, Undergraduate Students
University of Arizona, Tucson, AZ
- 2014 - 2016 *Tucson Women in Astronomy Mentorship Program Mentor*, Undergraduate Students
University of Arizona, Tucson, AZ
- 2014 - 2017: *Steward Observatory Graduate-Postdoc Mentoring Program Mentor and Mentee*, Junior Graduate Students
- 2013 - 2014: *Huffington Post Girls in STEM Mentorship Program Mentee*
New York University, New York, NY

Professional Development

- 2014: AAS/DPS Astronomy Ambassador Workshop, Tucson, AZ

TEACHING EXPERIENCE

- Spring 2018: *Teaching Assistant*, ASTR400B, Astronomy Tutoring for Majors & Minors Program
University of Arizona, Tucson, AZ
- Spring 2017: *Teaching Assistant*, Astronomy Tutoring for Majors & Minors Program
University of Arizona, Tucson, AZ
- 2015 - 2017: *Mathematics Tutor for Elementary through High School Students*, Ph.G. Tutoring
Tucson, AZ
- Summer 2014: *Laboratory Teaching Assistant*, Einstein's Universe (non-majors)
New York University, New York, NY
- Spring 2014: *Recitation Teaching Assistant*, Physics II (majors)
New York University, New York, NY
- Summer 2013: *Physics Adjunct*, General Physics I (non-majors)
New York University, New York, NY

PUBLICATIONS LIST

8 journal articles (5 first or second author refereed articles, 1 in preparation)

8. *The Effect of the Large Magellanic Cloud on the Orbits of Milky Way Dwarf Galaxies Using Gaia DR2*
Patel, E. et al., in prep. (expected submission: December 2018)
7. *ΛCDM Predictions for the Satellite Population of M33*
Patel, E., Carlin, J., Tollerud, E., Collins, M., Dooley, G., 2018, MNRAS, 480, 1883-1897
6. *First Gaia Dynamics of the Andromeda System: DR2 Proper Motions, Orbits, and Rotation of M31 and M33*
van der Marel, R. P., Fardal, M., Sohn, S.T., **Patel, E.**, et al., 2018, ApJ, in press (arXiv:1805.04079)
5. *The Frequency of Dwarf Galaxy Multiples at Low Redshift in SDSS vs. Cosmological Expectations*
Besla, G., Patton, D., Stierwalt, S., Rodriguez-Gomez, V., **Patel, E.**, et al., 2018, MNRAS. 480, 3376-3396
4. *Estimating the Mass of the Milky Way Using the Ensemble of Classical Satellite Galaxies*
Patel, E., Besla, G., Mandel, K., Sohn, S.T., 2018, ApJ, 857, 78-94
3. *Space Motions of the Dwarf Spheroidal Galaxies Draco and Sculptor Based on HST Proper Motions with ~ 10 Year Base-Line*
Sohn, S.T., **Patel, E.**, et al., 2017, ApJ, 849, 93-107
2. *The Orbits of Massive Satellite Galaxies - II. Bayesian Estimates of the Milky Way and Andromeda masses using high precision astrometry and cosmological simulations*
Patel, E., Besla, G., Mandel, K., 2017, MNRAS, 468, 3428-3449
1. *The Orbits of Massive Satellite Galaxies - I. A Close Look at the Large Magellanic Cloud and a New Orbital History for M33*
Patel, E., Besla, G., Sohn, S.T., 2017, MNRAS, 464, 3825-3849

CONFERENCE PROCEEDINGS

2. *Estimating the Mass of the Milky Way Using the Ensemble of Classical Satellite Galaxies*
Patel, E., Besla, G., Sohn, S. T., Mandel, K., 2018, American Astronomical Society, AAS Meeting 232, id. 402.03
1. *The Dynamics of the Local Group in the Era of Precision Astrometry*
Besla, G., Garavito-Camargo, N., **Patel, E.**, 2018, American Astronomical Society, AAS Meeting 232, id. 207.01

SELECTED PRESS RELEASES

- [University of Arizona Press Release](#): How do you Weigh a Galaxy? Especially the One You're In?
- [International Business Times](#): Milky Way's Mass Estimated More Reliably Using Satellite Galaxies Angular Momentum
- [Nature Research Highlights](#): Measuring the Milky Way's mind-boggling mass
- [AAS Nova](#): Using Satellite Galaxies to Weigh the Milky Way
- [LiveScience](#): How Massive is the Milky Way?
- [Space.com](#): Milky Weigh: New Method Pins Down Our Galaxy's Mass
- [Astronomy Magazine](#): A whole new way to weigh the Milky Way